

Step 1: Understand the Working of Medical Billing Software for Architecture Creation



Medical billing software is integral to the healthcare revenue cycle. This process involves multiple steps, including:

Patient Insurance Validation

This step verifies the validity of the patient's insurance coverage. The front desk operators carry it out before the appointment schedule. In this, information about the patient including medical history with diagnosis, number of total appointments, previous billing information, and more is validated by the software.

Charge Capture Records

These are the medical services provided to the patients and charges for the same. The doctors and their teams are responsible for filling out these details.

Medical Coding

Medical coding covers medical services, treatment procedures, diagnoses, and prescriptions into standard healthcare codes. These codes must be correctly determined based on the International Classification of Diseases (ICD-10),

current procedure terminology (CPT), and Healthcare Common Procedure Coding System (HCPCS) standards.

Claims Submission

This step audits and submits billable fees to the insurance companies for reimbursement.

A/R Follow Up

Account receivable follow-up ensures that the healthcare providers can recover overdue payments. It involves tracking denied claims, checking partial payments, re-filing claims, and represents the amount that patients owe to the hospital for services provided.

Denial Management

Denial management involves standard denial reason codes and highlights the gaps in billings, registration, and medical coding process. It also records this data to avoid future denials and ensure acceptance.

Remittance Processing

This refers to the inspection of payments resulting from a bill for rendered medical services to determine the acceptance or denial of payments.

Third-party Follow-Up

In this step, payments from the third-party are collected. This is an essential step as healthcare organizations can only choose Medicaid or Medicare after other options have failed.

Patient Collections

Patient collections collect all the co-payments and non-covered charges from patients if the bill for healthcare services is partially insured.



Reporting and Data Analytics

All the information is stored in the system for future use and analyzed to improve financial operations by extracting insights.

Step 2: Finalize the Required Features for your Medical Billing Software with You

1. Patient Preregistration

Optimized billing workflow is crucial for the accuracy of medical billing data. While it depends on multiple factors, consistency of patient pre-registration is one of them. This feature is also essential to foster better patient care and reduce waiting times. It collects the following data: Optimized billing workflow is crucial to the accuracy of medical billing data. While it depends on multiple factors, consistency of patient preregistration is one of them. This feature is also essential to foster better patient care and reduce waiting times. It collects the following data:

- Patient medical history
- Insurance eligibility
- Medicare provider enrollment
- Patient's details

If the preliminary information is available at the time of registration, it is easier to address gaps in the future, and also, the information is displayed to the doctors in concern in a timely and accurate manner. If the preliminary information is available at the time of registration, it is easier to address gaps in the future, and also, the information is displayed to the doctors in concern in a timely and accurate manner.

2. Claims Management

Claims processing for medical billing and coding is carried out through the credentialing feature. Claims management is one of the essential features of any medical billing software as it automates billing, saves time, and reduces inaccuracies. This includes other sub-tasks and corresponding features for:



Claim Process, Invoice creation, Claim Submission, Claim Update

- Treatment details
- Appointment type
- Revisits

An easy-to-use UI/UX that can be a common platform for scheduling and billing is ideal for medical billing software.

7. Compliance

As explained earlier, compliance is an essential requirement when it comes to healthcare software of any kind. Without complying with the standards, you risk running into legal issues that might permanently ban your software solution. Billing software with compliance will take your software to the zenith of success.

8. Reporting and Analysis

To generate medical billing services, financial reports, patient documentation, claims management, and accurate reporting and analysis are vital. Ensure your medical billing software allows seamless workflow between the various nodes to gather information and insights required. Use an easy-to-use dashboard that offers visualization for the user to derive results.

Step 3: Technology Stack that Would Best Suit Your Project Requirements

Once we know what feature we have to integrate in your solution, it is time to decide the technology stack. The goal behind this process is to pick the right technologies to deliver the best possible solution for your organization and its unique needs. Furthermore, many organizations are going beyond norms and investing in next-gen integration as well. Matellio provides IoT development services (<https://www.matellio.com/solutions/iot-development-services/>), AI development services, chatbot integration, and many other services that can truly make your billion software feel next-gen.

Here is the technology stack that our experts usually suggest:



Front End Development	
Libraries	Vue.js, React
Languages	CSS 3, JS, HTML 5
Back End Development	
Server	Nginx
App Frameworks	Laravel
Databases	My SQL, Redis
Hosting	AWS, VULTR

Step 4: Finalize the Team Structure

This brings us to an important step, helping you connect with the right people who will finally build your solution. Matellio houses a vast pool of talented developers, designers, and engineers and we make sure that we allot the right talent to the right project,

Your team shall typically include:

- Project manager
- Front-end developer
- Back-end developer
- UI/UX experts
- QA and testing engineers

Step 5: Custom Medical Billing Software Development and QA/Testing



Once everything is finalized, we'll start working on the project to deliver the expected results. This, too, is broken down into multiple stages such as:

- **SRS Creation:** A software requirement specification (SRS) document is a vital first step towards developing any software. It captures the complete project description, including how a system shall operate and other technicalities. In this stage, a detailed view of the system is designed on paper, with each aspect explained in detail. Once the client approves the SRS document, the following phases of the document shall begin. Developers will consider the requirements mentioned here as benchmarks, so the client must ensure that all the requirements are specified.
- **Software Development:** Based on the SRS, the development work will start. It is mainly carried out in UI/UX development, front-end development, and back-end development. The project manager oversees the entire process to ensure cutting-edge delivery and shall stay connected to you to share regular updates.
- **QA/ Testing:** QA and testing experts design and run the system based on various use cases to ensure that the system works like it is intended to. This is a crucial stage to deliver software that not just meets your requirements but is also bug-free.

coverage, and revenue cycle management. Automated accounting helps streamline the entire process with features that include:

- E-Invoices
- Payment Reminders
- Coding
- Claims management

4. Data Exchange and Collaboration

Throughout the cycle of medical billing, including invoicing, coding, appointment scheduling, data integration, or data sharing among various stakeholders, medical billing software should encompass all smooth data updates to ensure each user receives the latest copy of the data. Even a single gap in the data can lead to big mistakes that can be cost the organization handsomely.

5. Patient Data

A centralized patient data repository helps healthcare organizations to stay updated. It helps medical billing systems and other such software to collaborate seamlessly. When provided with accurate patient information, the medical billing process can be instant and uninterrupted. For this, the medical billing systems need to collect up-to-the-minute information from EHR/ EMR along with the following features:

- Medical records
- Insurance eligibility
- Medicare coverage
- Prescription history

6. Appointment Scheduling

Patient appointment scheduling helps medical billing software to ensure accuracy in medical billing codes. The following features integrated into the application will help with simplified medical billing:

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- Patient inflow